Title : Analyzing Sales Performance by Region in a Retail Company

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

In [3]:
```
df = pd.read_csv("./datasets/customer_shopping_data.csv")
df.head()
```

Out[3]:

	invoice_no	customer_id	gender	age	category	quantity	price	payment_method	invoice_da
0	I138884	C241288	Female	28	Clothing	5	1500.40	Credit Card	5/8/20
1	I317333	C111565	Male	21	Shoes	3	1800.51	Debit Card	12/12/20
2	I127801	C266599	Male	20	Clothing	1	300.08	Cash	9/11/20
3	I173702	C988172	Female	66	Shoes	5	3000.85	Credit Card	16/05/20
4	I337046	C189076	Female	53	Books	4	60.60	Cash	24/10/20

In [4]:
```
df.tail()
```

Out[4]:

	invoice_no	customer_id	gender	age	category	quantity	price	payment_method	inv
99452	I219422	C441542	Female	45	Souvenir	5	58.65	Credit Card	21
99453	I325143	C569580	Male	27	Food & Beverage	2	10.46	Cash	22
99454	I824010	C103292	Male	63	Food & Beverage	2	10.46	Debit Card	28
99455	I702964	C800631	Male	56	Technology	4	4200.00	Cash	16
99456	I232867	C273973	Female	36	Souvenir	3	35.19	Credit Card	15

In [6]:
```
# To check the count of records grouped by region/branch of the mall
df.groupby("shopping_mall").count()
```

Out[6]:

shopping_mall	invoice_no	customer_id	gender	age	category	quantity	price	payment_metho
Cevahir AVM	4991	4991	4991	4991	4991	4991	4991	499
Emaar Square Mall	4811	4811	4811	4811	4811	4811	4811	481
Forum Istanbul	4947	4947	4947	4947	4947	4947	4947	494
Istinye Park	9781	9781	9781	9781	9781	9781	9781	978
Kanyon	19823	19823	19823	19823	19823	19823	19823	1982
Mall of Istanbul	19943	19943	19943	19943	19943	19943	19943	1994
Metrocity	15011	15011	15011	15011	15011	15011	15011	1501
Metropol AVM	10161	10161	10161	10161	10161	10161	10161	1016
Viaport Outlet	4914	4914	4914	4914	4914	4914	4914	491
Zorlu Center	5075	5075	5075	5075	5075	5075	5075	507

In [8]:
```
# To check the count of records grouped by the product categories
df.groupby("category").count()
```

Out[8]:

category	invoice_no	customer_id	gender	age	quantity	price	payment_method	invoice_da
Books	4981	4981	4981	4981	4981	4981	4981	49
Clothing	34487	34487	34487	34487	34487	34487	34487	344
Cosmetics	15097	15097	15097	15097	15097	15097	15097	150
Food & Beverage	14776	14776	14776	14776	14776	14776	14776	147
Shoes	10034	10034	10034	10034	10034	10034	10034	100
Souvenir	4999	4999	4999	4999	4999	4999	4999	49
Technology	4996	4996	4996	4996	4996	4996	4996	49
Toys	10087	10087	10087	10087	10087	10087	10087	100

In [9]:
```
# total sales for each mall branch
branch_sales = df.groupby("shopping_mall").sum()
branch_sales
```

Out[9]:

shopping_mall	age	quantity	price
Cevahir AVM	215474	14949	3433671.84
Emaar Square Mall	209575	14501	3390408.31
Forum Istanbul	215380	14852	3336073.82
Istinye Park	424335	29465	6717077.54
Kanyon	862280	59457	13710755.24
Mall of Istanbul	866333	60114	13851737.62
Metrocity	652968	44894	10249980.07
Metropol AVM	439086	30530	6937992.99
Viaport Outlet	212771	14716	3414019.46
Zorlu Center	220926	15234	3509649.02

In [11]:
```
# total sales for each category of product
category_sales = df.groupby("category").sum()
category_sales
```

Out[11]:

category	age	quantity	price
Books	216882	14982	226977.30
Clothing	1497054	103558	31075684.64
Cosmetics	657937	45465	1848606.90
Food & Beverage	640605	44277	231568.71
Shoes	436027	30217	18135336.89
Souvenir	216922	14871	174436.83
Technology	216669	15021	15772050.00
Toys	437032	30321	1086704.64

In [12]:
```
# to get the top performing branches
branch_sales.sort_values(by = "price", ascending = False)
```

Out[12]:

shopping_mall	age	quantity	price
Mall of Istanbul	866333	60114	13851737.62
Kanyon	862280	59457	13710755.24
Metrocity	652968	44894	10249980.07
Metropol AVM	439086	30530	6937992.99
Istinye Park	424335	29465	6717077.54
Zorlu Center	220926	15234	3509649.02
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Viaport Outlet	212771	14716	3414019.46
Emaar Square Mall	209575	14501	3390408.31
Forum Istanbul	215380	14852	3336073.82

In [13]:
```
# to get the top selling categories
category_sales.sort_values(by = "price", ascending = False)
```

Out[13]:

category	age	quantity	price
Clothing	1497054	103558	31075684.64
Shoes	436027	30217	18135336.89
Technology	216669	15021	15772050.00
Cosmetics	657937	45465	1848606.90
Toys	437032	30321	1086704.64
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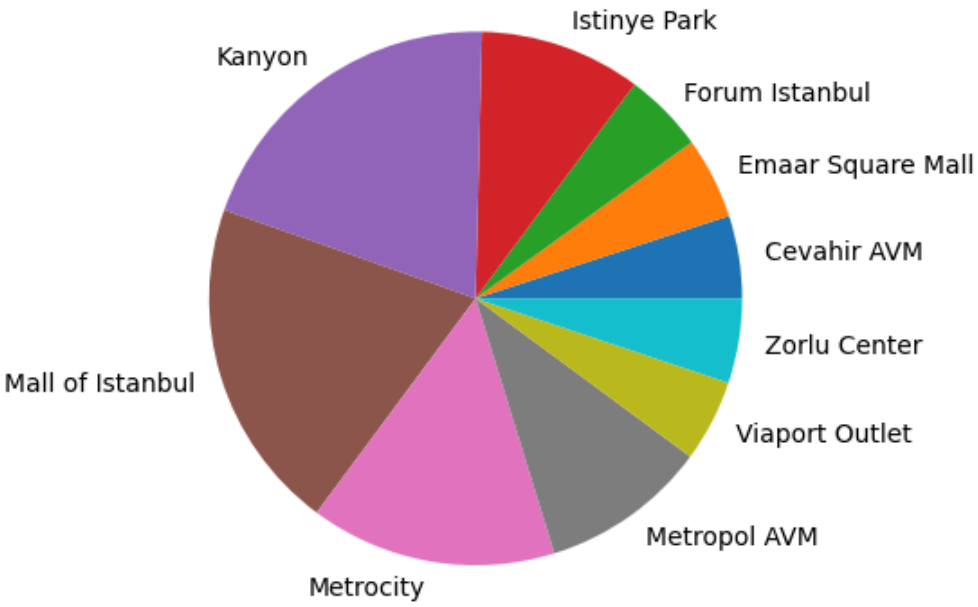
In [15]:
```
# to get total sales for each combination of branch and product_category
combined_branch_category_sales = df.groupby(["shopping_mall", "category"]).sum()
combined_branch_category_sales
```

Out[15]:

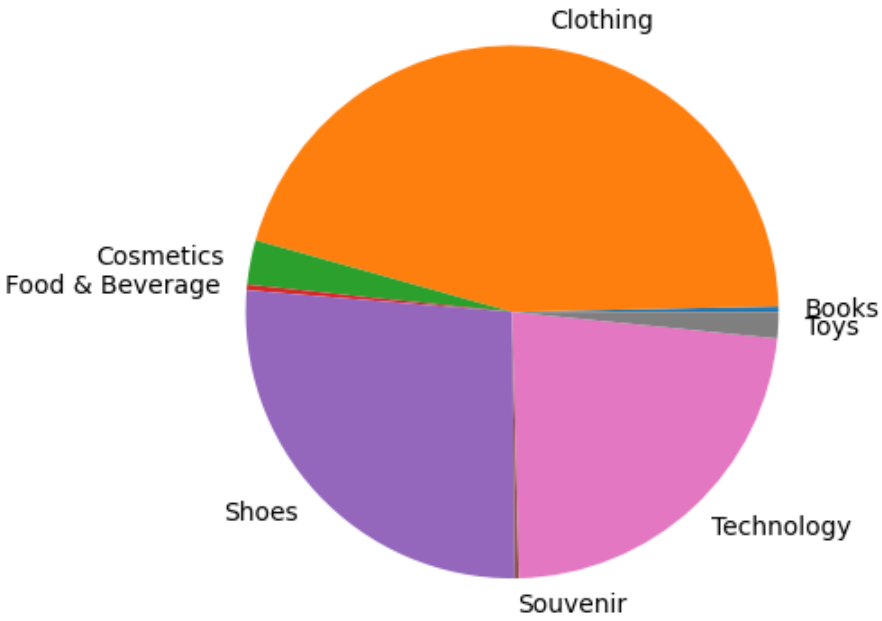
shopping_mall	category	age	quantity	price
Cevahir AVM	**Books**	11464	792	11998.80
	Clothing	74729	5180	1554414.40
	Cosmetics	31142	2174	88394.84
	Food & Beverage	33269	2293	11992.39
	Shoes	21211	1473	884050.41
...	**...**	...	...	...
Zorlu Center	**Food & Beverage**	32687	2216	11589.68
	Shoes	22949	1589	953670.13
	Souvenir	10727	716	8398.68
	Technology	10533	765	803250.00
	Toys	22395	1526	54691.84

80 rows × 3 columns

In [16]:
```
# pie chart for sales by branch
plt.pie(branch_sales["price"], labels = branch_sales.index)
plt.show()
```

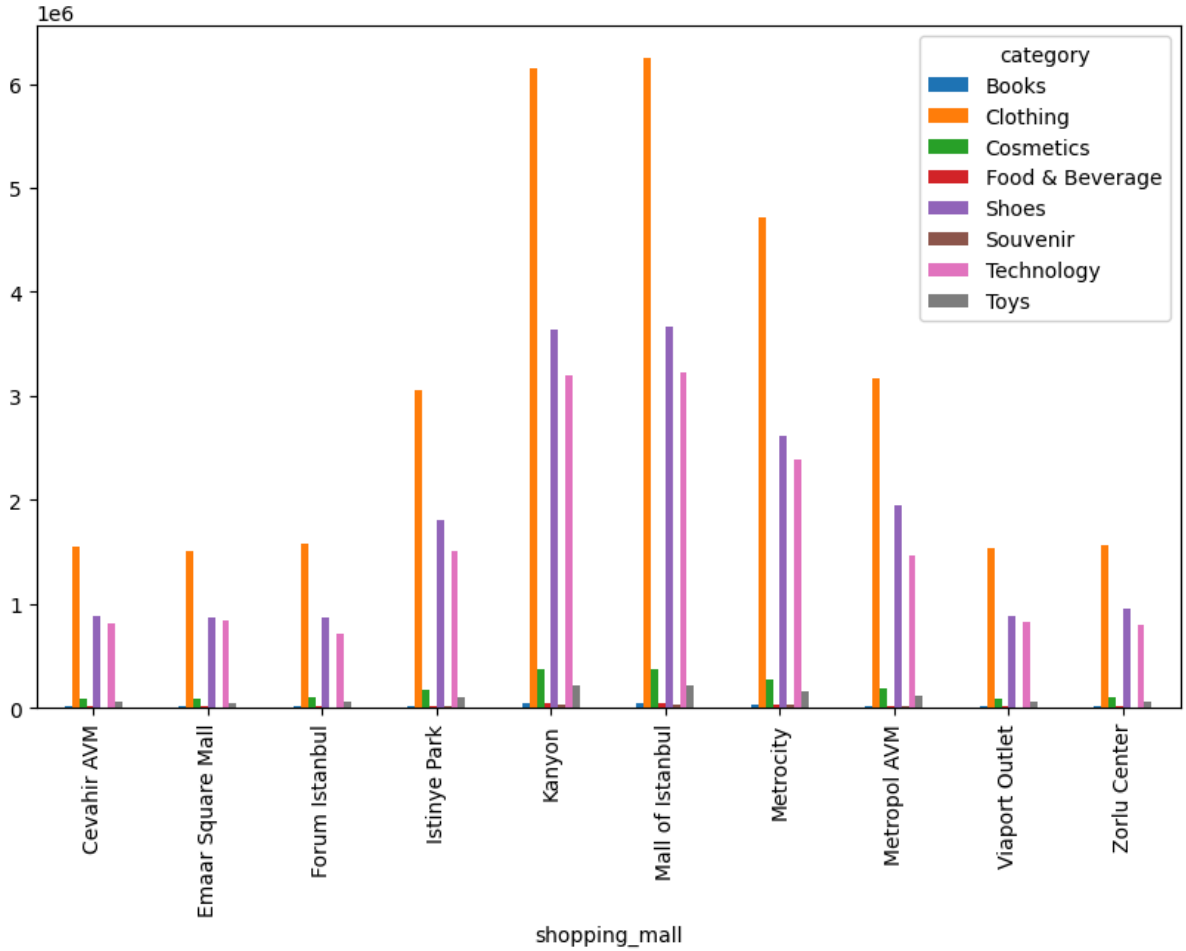


In [17]:
```
# pie chart for sales by product category
plt.pie(category_sales["price"], labels = category_sales.index)
plt.show()
```



In [19]:
```
combined_pivot = df.pivot_table(index="shopping_mall", columns="category", values="price",
```

In [20]:
```
# grouped bar chart for sales of different categories at different branches
combined_pivot.plot(kind="bar", figsize=(10, 6))
plt.show()
```



In [ ]:
```

```